C14 # Agent Harness Map

Design Advanced

Choose a runtime, repository contract, or orchestrator

The four layers

Layer	Responsibility	Does not replace
Model	Proposes text and tool calls	Permission and recovery
Runtime harness	Runs the loop and tools	Repository contract
Repository harness	Instructions, state, verification	Runtime loop
Orchestrator	Coordinates runtimes and sessions	Each runtime's decisions

An orchestrator can launch several runtimes without owning their loops. A repository can improve its contract without changing runtime.

Loop and graph are not extra layers. Loop names the repeated feedback and stopping rule. Graph names the executable nodes, branches, joins, and checkpoints inside a runtime or an orchestrator.

A harness optimizer sits outside this stack. It changes and evaluates candidate harnesses for later runs.

Choose in the right order

1. Define the task and acceptable autonomy.
2. Select a runtime that owns the required loop.
3. If the flow must be versioned, specify nodes, branches, and checkpoints inside the chosen runtime.
4. Write the repository contract: instructions, pass criteria, verification.
5. Add an orchestrator only when coordination, isolation, or recovery across tasks requires it.

Five sources, five uses

Source	Used to	Limit
Strict comparison	Identify runtimes	Not a universal ranking
Orchestrator table	Compare coordination	Not a runtime
Best of Agent Harnesses	Explore 160 projects, 12 categories	2026-08-23 snapshot
`agent-harnesses.json`	Read provenance and evidence	`unknown` is not `none`
Optimizer research table	Compare search protocols	Mostly research systems

Statuses for observed mechanisms: `confirmed` = visible mechanism, `claimed` = source assertion, `unknown` = not established yet, `not_applicable` = category mismatch.

Short trial, traceable verdict

Compare two or three model-harness pairs on real tasks in separate worktrees. Repeat critical tasks. Measure human verdict, interventions, plan drift, wall time, accepted-task cost, recovery, and setup friction.

Links: Agent Harness Map · Loop & Graph Engineering · Agent Harness Engineering · Agent Tools · Glossary